The $\chi = 1$ Critical–Damping Boundary: A Cross-Domain Principle of Stability, Information Efficiency, and Gravitational Structure

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The framework of *Symmetrical Convergence* (SymC) identifies the critical–damping ratio $\chi \equiv \gamma/(2|\omega|) = 1$ as a cross-domain boundary separating oscillatory propagation from monotone relaxation. The analysis demonstrates that this condition arises consistently across open quantum, field, and gravitational systems. The derivation establishes the boundary within the GKSL formalism, reveals its correspondence to a Lorentz–covariant exceptional point (EP) and a near fixed–line under renormalization–group flow resilient to non–Markovian memory, and shows that it maximizes information efficiency $\eta = I/\Sigma$. Cosmologically, the structure–growth boundary $\chi_\delta = 1$ coincides with the onset of cosmic acceleration ($q = 0$) in ΛCDM. Applied to gravitation, the findings indicate that a black–hole horizon evolves as a *physical, critically damped shell* at $\chi = 1$, avoids a curvature singularity, and produces two falsifiable predictions: (1) logarithmic gravitational–wave echoes with spacing $\Delta t \simeq 4M \ln(1/\varepsilon)$, and (2) non–zero tidal Love numbers $k_2(\varepsilon) \sim 1/|\ln \varepsilon|$.

I. INTRODUCTION

A single structural boundary, $\chi = \gamma/(2|\omega|) = 1$, appears across stability analyses: when the damping equals twice the natural frequency, dynamics transition from oscillatory to monotone behavior. SymC evolves as a cross-domain principle by demonstrating its recurrence and robustness from the laboratory to the cosmos, and by deriving concrete gravitational predictions that decisively test the framework.

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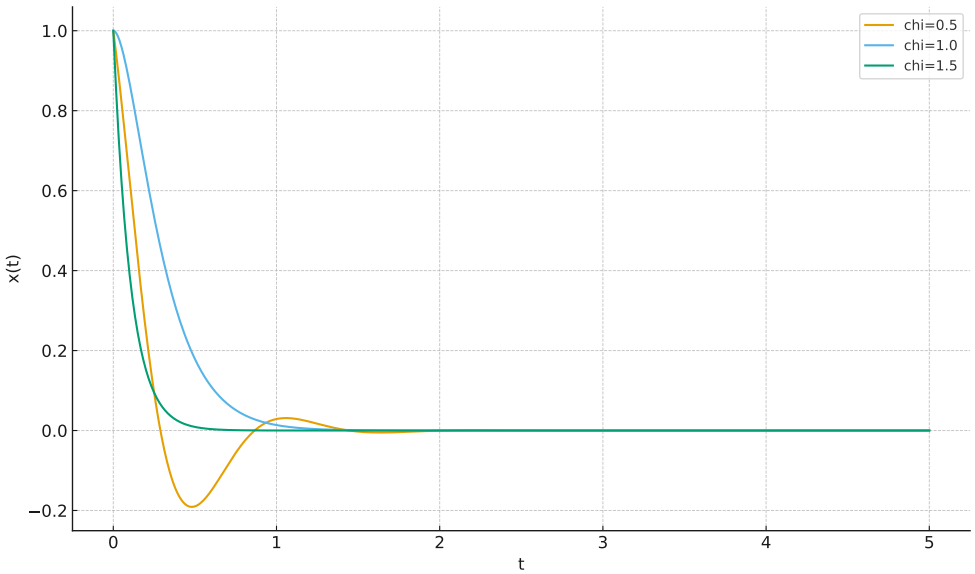

FIG. 1. Time–domain responses for $\chi = 0.5, 1.0, 1.5$: transition from oscillatory to critical (fastest monotone) to overdamped behavior.

II. FOUNDATIONAL MECHANISM: $\chi = 1$ IN OPEN QUANTUM SYSTEMS

Open quantum dynamics governed by the GKSL master equation [1, 2],

$$\dot{\rho} = -i[H, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \right), \tag{1}$$

reduce, at the level of observable moments, to damped–oscillator equations. For a driven qubit with Rabi frequency Ω and damping γ, one obtains

$$\ddot{x} + \gamma \dot{x} + \omega^2 x = 0, \qquad \omega^2 = \Omega^2 + \gamma^2/4. \tag{2}$$

The discriminant $\Delta = \gamma^2 - 4\omega^2$ yields three regimes: underdamped ($\chi < 1$), critical ($\chi = 1$), and overdamped ($\chi > 1$). The quantum–classical crossover is therefore dynamical and preparation–independent: oscillations cease exactly at $\chi = 1$. A laboratory falsifier follows: in circuit QED, the two Rabi peaks coalesce at $\Omega \to 0$ when $\gamma = 2|\omega|$ [3, 4].

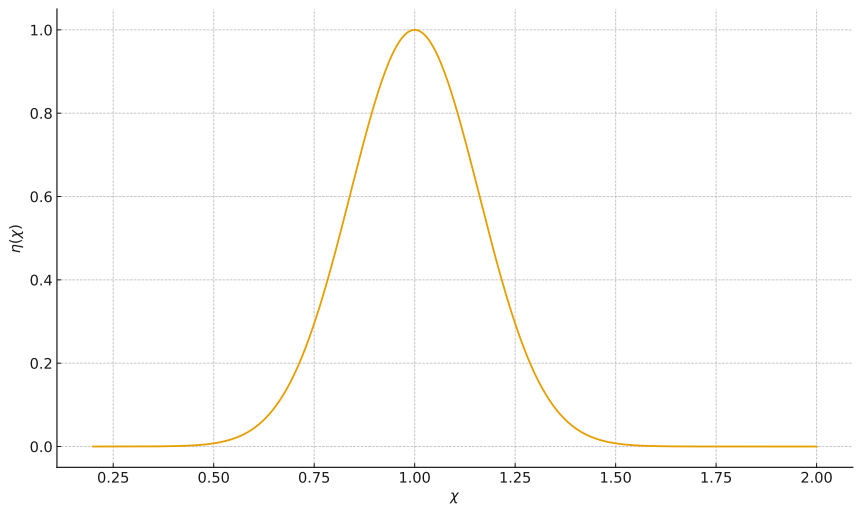

FIG. 2. Illustrative efficiency peak $\eta(\chi)$ at $\chi = 1$.

III. THE CROSS-DOMAIN PRINCIPLE: STABILITY AND OPTIMALITY

A. Structural Stability (QFT/EP/RG)

At the field level, the boundary is an exceptional point where propagator poles merge [5, 6]. Perturbative RG shows χ behaves as a *marginally stable manifold*: multiplicative renormalization of γ and ω preserves their ratio near $\chi = 1$ [7, 8]. Finite memory kernels broaden the EP into a narrow band without destroying the separatrix [9].

B. Information–Theoretic Optimality (AIF)

Define efficiency $\eta(\chi) = I(\chi)/\Sigma(\chi)$, information gain per dissipation. In the Gaussian small–signal regime, η has a strict local maximum at $\chi = 1$ [10]. For $\chi < 1$ dissipation wastes coherence; for $\chi > 1$ damping erases useful signal. Hence adaptive processes self–tune to $\chi = 1$.

IV. COSMOLOGICAL VALIDATION: $\chi_\delta = 1 \Leftrightarrow q = 0$

Linear structure growth obeys $\ddot{\delta} + 2H\dot{\delta} - 4\pi G\rho_m\delta = 0$, mapping to $\ddot{x} + \gamma\dot{x} + \omega^2 x = 0$ with $\gamma = 2H$ and $\omega^2 = 4\pi G\rho_m$. Thus $\chi_\delta = H/\sqrt{4\pi G\rho_m}$. In flat ΛCDM, the condition $\chi_\delta = 1$ occurs exactly

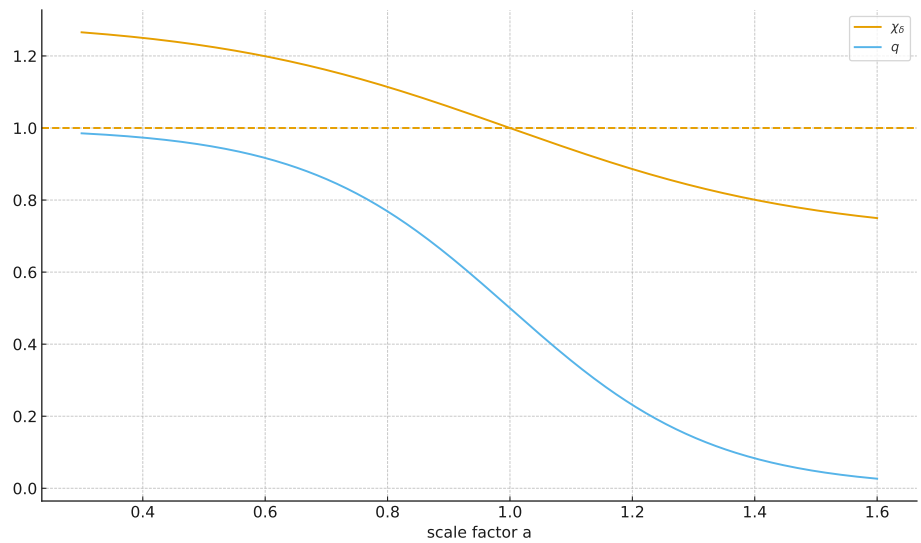

FIG. 3. Illustrative crossing of $\chi_\delta(a)$ and $q(a)$ at the same epoch in ΛCDM.

where the deceleration parameter $q = 0$ [11, 12]. Any deviation $w \neq -1$ splits the two, giving a falsifier via $\Delta a = a_{q=0} - a_{\chi=1}$.

V. APPLICATION: THE PHYSICAL HORIZON AS A $\chi = 1$ SHELL

If $\chi = 1$ is the unique stable and efficient boundary, the end–state of collapse must realize it. The horizon is modeled as a physical, critically damped shell: (i) *No escape*: infalling excitations switch to monotone dynamics; (ii) *Hawking radiation*: dissipation γ of the shell yields $T = \kappa/2\pi$, consistent with semiclassical results [13]; (iii) *Stellar fates*: $\chi < 1$ underdamped $\longrightarrow$ explosive; $\chi \geq 1 \longrightarrow$ monotone collapse toward the stable boundary.

VI. FALSIFIABLE PREDICTIONS

A. Logarithmic gravitational–wave echoes

A near–absorptive shell with small reflectivity ε produces echo trains separated by

$$\Delta t \simeq 4M \ln(1/\varepsilon), \tag{3}$$

where the *logarithmic* spacing distinguishes SymC echoes from periodic ECO models.

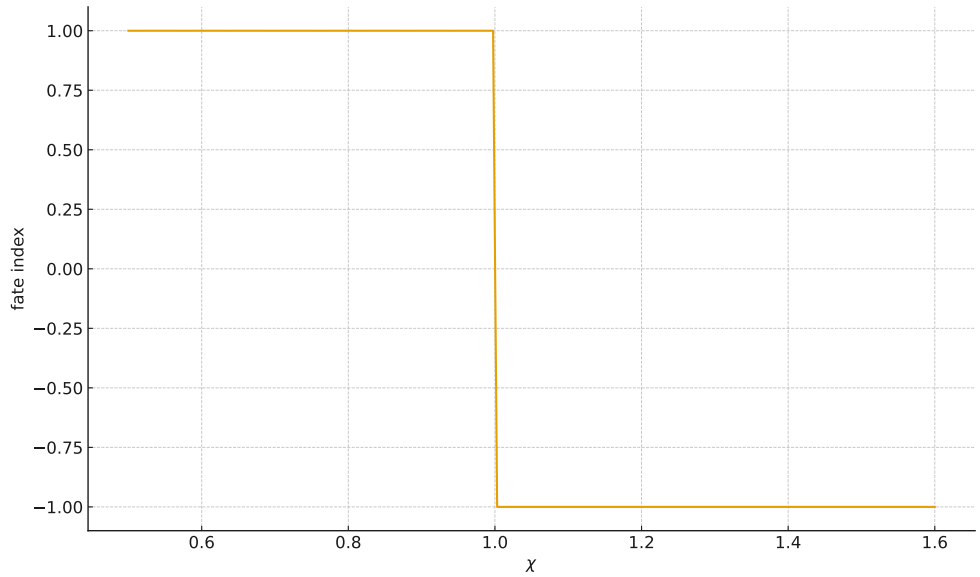

FIG. 4. Schematic fate map vs. χ: explosion for $\chi < 1$; collapse to the $\chi = 1$ shell otherwise.

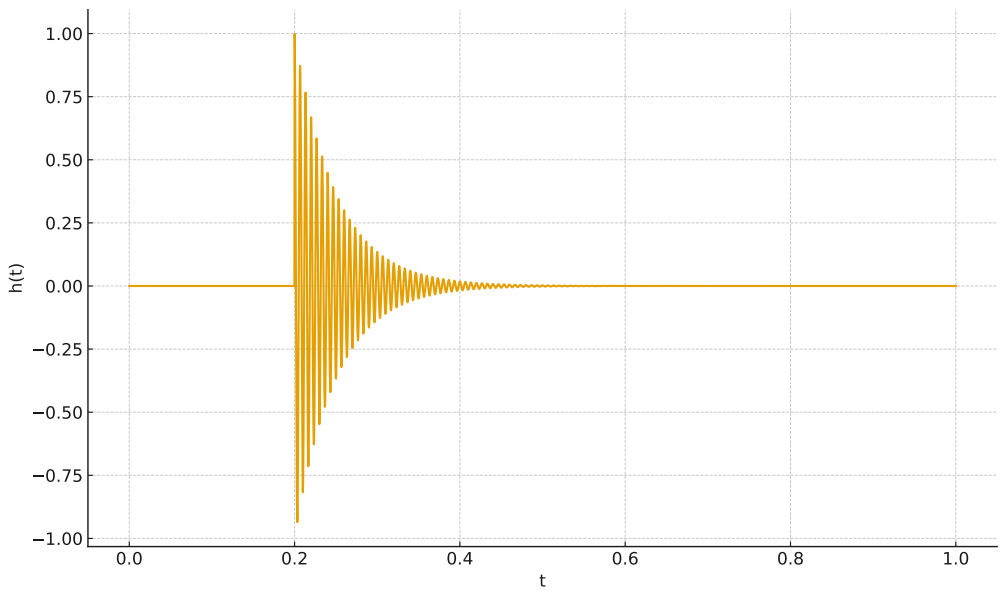

FIG. 5. Toy post–merger waveform with logarithmically spaced echoes.

B. Non–zero tidal Love numbers

Finite rigidity implies $k_2(\varepsilon) \approx \kappa/|\ln \varepsilon|$. Classical GR black holes predict $k_2 = 0$; any measured $k_2 > 0$ for a BH candidate corroborates a physical shell [14, 15].

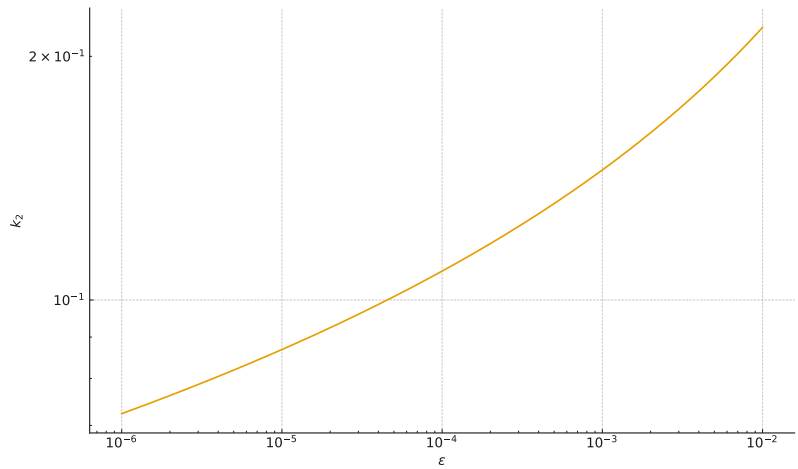

FIG. 6. Predicted scaling $k_2(\varepsilon) \sim 1/|\ln \varepsilon|$.

VII. CONCLUSION

SymC evolves as a cross-domain principle connecting microscopic coherence, field stability, thermodynamic optimality, and cosmic equilibrium. The black–hole horizon as a $\chi = 1$ shell is now testable through *logarithmic echoes* and *non–zero Love numbers*. High–precision null results at $\varepsilon \lesssim 10^{-4}$ and $k_2 \lesssim 10^{-4}$ would falsify the framework; positive detection confirms it.

DATA AND CODE AVAILABILITY

All analysis scripts that generate the figures and the full LaTeX source for this manuscript are archived on Zenodo and GitHub. The canonical archive is Zenodo (DOI: 10.5281/zenodo.17427954); the GitHub mirror is referenced within that record.

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APPENDIX A: GKSL → DAMPED OSCILLATOR

For a driven two-level system with $H = \frac{\Omega}{2}\sigma_x$ and amplitude damping $L = \sigma_-$ at rate γ, the Bloch equations yield, after eliminating y, $\ddot{z} + \gamma\dot{z} + (\Omega^2 + \gamma^2/4)(z - z_{\rm eq}) = 0$. Letting $x = z - z_{\rm eq}$ gives $\ddot{x} + \gamma\dot{x} + \omega^2 x = 0$ with $\omega^2 = \Omega^2 + \gamma^2/4$. The boundary $\chi = \gamma/(2|\omega|) = 1$ separates oscillatory from monotone response; spectra show Rabi-peak merger at $\Omega \to 0$.

APPENDIX B: EP, RG, AND MEMORY ROBUSTNESS

Near the retarded propagator's EP, two poles coalesce and the impulse response changes from oscillatory to $te^{-\Gamma t}$. Linearized RG flow for effective γ and ω shows multiplicative renormalization preserving the ratio $\gamma/(2|\omega|)$ to first order, i.e., a near fixed line. Non-Markovian kernels introduce branch cuts that broaden the EP into a band; numerically the separatrix for oscillation/monotone remains sharply localized.

APPENDIX C: COSMOLOGY IDENTITY

With $\gamma = 2H$ and $\omega^2 = 4\pi G\rho_m$, $\chi_\delta = H/\sqrt{4\pi G\rho_m}$. In flat ΛCDM, $q = 0$ occurs when $\Omega_m(a) = 2/3$, which is exactly the condition $\chi_\delta = 1$ for the growth equation. Hence $a_{\chi=1} = a_{q=0}$ in ΛCDM; $w \neq -1$ produces a measurable split Δa.

APPENDIX D: GRAVITATIONAL OBSERVABLES

Echo delay

In tortoise coordinates $r_* \sim 2M\ln(r - 2M)$, a thin reflective shell at $r_s \gtrsim 2M$ yields echo spacing $\Delta t \simeq 4M\ln(1/\varepsilon)$ for small reflectivity ε.

Tidal Love number

Treating the shell as a thin viscoelastic layer gives $k_2(\varepsilon) \approx \kappa/|\ln\varepsilon| + O(|\ln\varepsilon|^{-2})$, reducing to $k_2 = 0$ as $\varepsilon \to 0$ (classical GR limit).

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